

SQL bootcamp by techTFQ

▼ Course Contents

1. Introduction & Pre-requisites

- a. Course Introduction
 - i. Welcome message
 - ii. What will you learn in this course?
 - iii. Who is this course for?
- b. Introduction to SQL
 - i. What is Data?
 - ii. What is Database?
 - iii. How is data stored in database?
 - iv. What sort of database are we using?
 - v. What is SQL?
 - vi. Why do we use SQL?
 - vii. What is a Schema?
- c. Installing database and IDE
 - i. Which database should you choose? This course is applicable to which database?
 - ii. Install PostgreSQL in Mac and Windows
 - iii. Install PgAdmin too in Mac and Windows
- d. Exercise/Practice
 - a. Create new database & schema

2. Basic SQL

- a. SQL Commands

- i. DDL (Data Definition Language)
 - 1. CREATE
 - 2. DROP
 - 3. ALTER
 - 4. TRUNCATE
 - ii. DML (Data Manipulation Language)
 - 1. INSERT
 - 2. UPDATE
 - 3. DELETE
 - 4. MERGE
 - iii. DCL (Data Control Language)
 - 1. GRANT
 - 2. REVOKE
 - iv. TCL (Transaction Control Language)
 - 1. COMMIT
 - 2. ROLLBACK
 - 3. SAVEPOINT
 - v. DQL (Data Query Language)
 - 1. SELECT
- b. Data Types
- i. INT
 - ii. VARCHAR (TEXT)
 - iii. DATE
 - iv. FLOAT (DECIMAL)
 - v. BOOLEAN
 - vi. IDENTITY column (Auto Increment)

c. Constraints

- i. Primary Key
- ii. Foreign Key
- iii. Unique
- iv. Check
- v. Not Null
- vi. Default

d. Exercise/Practice

e. Operators in SQL

- i. Arithmetic Operators (+ , - , / , * , %)
- ii. Logical Operators (AND , OR , NOT , EXISTS , IN , LIKE , ANY , ALL , BETWEEN , SOME)
- iii. Comparison Operators (= , <> , ! = , < , > , < = , > =)

f. CASE Statement

g. Important SQL Clause

- i. DISTINCT
- ii. ORDER BY
- iii. LIMIT (TOP)
- iv. UNION , UNION ALL

h. INNER Join

- i. CASE Study /Project - 1 (Solve 30 basic SQL problems)

3. **Intermediate SQL**

a. Group By & Having clause

b. Aggregate functions

- i. MIN , MAX , SUM , COUNT , AVG

c. Order of Execution

- d. Normalisation in SQL
 - i. 1NF
 - ii. 2NF
 - iii. 3NF
- e. CASE Study /Project - 2 (Solve 10 SQL problems)
- f. Sub-Queries
 - i. Scalar Subquery
 - ii. Multi-row subquery
 - iii. Correlated subquery
- g. CTE table (WITH clause)
- h. All types of JOINS
 - i. LEFT Join
 - ii. RIGHT Join
 - iii. Full Join
 - iv. Cross Join
 - v. SELF Join
 - vi. NATURAL Join
- i. CASE Study /Project - 3 (Solve 10 SQL problems)
- j. In-built functions in SQL
 - i. Date Functions
 - ii. String Functions
- k. CASE Study /Project - 4 (Solve 10 SQL problems)
- l. Window functions
 - i. RANK
 - ii. DENSE_RANK
 - iii. ROW_NUMBER

- iv. LEAD
- v. LAG
- vi. FIRST_VALUE
- vii. LAST_VALUE
- viii. NTH_VALUE
- ix. NTILE
- m. Views
- n. CASE Study /Project - 5 (Solve 20 SQL problems)

4. **Advanced SQL**

- a. Recursive SQL Queries
- b. PIVOT table, CROSSTAB function
- c. Materialized Views
- d. Stored Procedure
 - i. Variables
 - ii. Parameters
 - iii. IF Statement
 - iv. Exception handling
 - v. Loop statements
 - vi. Print statement
- e. User Defined functions
- f. CASE Study /Project - 6 (Solve 5 SQL problems)

5. **Interview Problems**

- a. 30 Interview problems
 - i. 20 easy
 - ii. 20 medium
 - iii. 10 hard